WORLD DUNGEON — MASTER TECHNICAL SPECIFICATION & IMPLEMENTATION MANUAL

1. EXECUTIVE SUMMARY & DESIGN PHILOSOPHY

1.1 Core Concept

World Dungeon is a hyperlocal, mobile-first Progressive Web App (PWA) and autonomous procedural quest-generation engine. It converts the player's real-world physical neighborhood into a fantasy dungeon quest crawl. Unlike fitness apps that treat walking as metric tracking (calories, paces, heart rate), World Dungeon frames every physical step as an occult exploration into forgotten local geography.

1.2 Aesthetic Directives

The application strictly rejects clean, modern, sterile mobile app aesthetics (e.g., standard flat design, rounded cards, standard Material/iOS navigation, bright saturated UI accents). Instead, it adopts a **dark fantasy, burnt manuscript, gothic tome** visual design inspired by *Elden Ring*, *Hollow Knight*, and *Hades*:

- **Color Palette Tokens**:
  - `dungeon-bg`: `#0a0a0f` (Near-black void backdrop)
  - `dungeon-gold`: `#c9a84c` (Aged gold for borders, accents, icons, and primary typography)
  - `dungeon-parchment`: `#e8e0d0` (Ancient parchment for body copy and readable typography)
  - `dungeon-burn`: `#1a0a00` (Deep burnt brown for redacted manuscripts)
  - `dungeon-ash`: `#3d2b1f` (Ash brown for burnt edges and progress tracks)
  - `dungeon-ember`: `#ff4500` (Ember orange for XP flash gains, level-ups, and burning boundaries)
- **Typography**:
  - Headings, labels, level badges, HUD chrome: **Cinzel** (Google Fonts).
  - Lore narrative, premises, waypoint titles, italicized excerpts: **IM Fell English** (Google Fonts).

- **Stylistic Constraints**:
    - Strict zero-border-radius across the entire application (sharp corners everywhere, with the sole exception of circular camera capture buttons).
    - No standard CSS box shadows; depth is achieved strictly through custom radial vignettes, SVG noise filters, and layered opacity masks.
    - No external UI component frameworks (No shadcn/ui, MUI, Chakra, or Tailwind UI). Everything is implemented from scratch using Tailwind CSS utility classes and custom inline SVG vectors.
    - Zero external raster icon sets; all glyphs, roses, and ornamental dividers are rendered inline via SVG.

2. REPOSITORY & ARCHITECTURE MAP

The project consists of two distinct, complementary systems:

1. **Frontend PWA**: Next.js 14 (App Router) with TypeScript, Tailwind CSS, Web Speech API, Geolocation, Canvas, and HTML5 file capture.
2. **Backend Engine**: Fastify (Node.js 18+), TypeScript, Zod, Overpass QL (OpenStreetMap), local Ollama LLM (`gemma2:2b`), Wikimedia Commons API, and SerpApi.

```
d:\World-dungeon\
├── .env.example                        # Frontend environment template
├── .env.local                          # Frontend local environment configuration
├── next.config.mjs                     # Next.js runtime configuration & image domains
├── package.json                        # Next.js dependencies and scripts
├── postcss.config.js                   # PostCSS with Tailwind & Autoprefixer
├── tailwind.config.ts                  # Dark fantasy color tokens & typography setup
├── tsconfig.json                       # Next.js TypeScript config (excludes backend)
├── public/
│   ├── icon.svg                        # PWA Gold Compass Rose Vector Icon
│   └── manifest.json                   # PWA Web App Manifest
├── src/
│   ├── app/
│   │   ├── globals.css                 # Keyframes, scanlines, shimmers, flash
animations
│   │   ├── layout.tsx                  # Root layout, Google Fonts injection, PWA
metadata
│   │   ├── page.tsx                    # Screen 1: The Living Breathing Home Screen
│   │   ├── walk/
│   │   │   └── page.tsx                # Screen 2: High-contrast 1-Second Glancable Walk
HUD
│   │   ├── artifact/
│   │   │   └── page.tsx                # Screen 3: Relic Vault, Map Exporter & Scrapbook
│   │   └── api/
│   │       ├── generate-quest/route.ts   # Gateway proxy to Fastify with archive fallback
│   │       ├── fetch-place-image/route.ts# Gateway proxy for Wikimedia place photography
│   │       ├── enrich/route.ts           # Gateway proxy for SerpApi landmark history
│   │       └── health/route.ts           # Gateway health check & status aggregator
│   ├── context/
│   │   └── QuestContext.tsx            # Central State Machine with localStorage
persistence
│   ├── hooks/
│   │   ├── useGeolocation.ts           # GPS watcher & 30s desktop interpolation engine
│   │   ├── useVoiceNarration.ts        # Web Speech API synthesized deep narrator voice
│   │   ├── useCamera.ts                # Direct mobile camera capture via HTML file
input
│   │   └── usePlayerLevel.ts           # Progression hooks & XP level calculation
│   ├── components/
│   │   ├── BurntManuscript.tsx         # Redacted burnt text reveal with SVG
feTurbulence
│   │   ├── CompassRose.tsx             # Hand-drawn gothic SVG compass with slow
rotation
│   │   ├── DungeonMapSVG.tsx           # 340x340 SVG parchment map with trace and
markers
```

```
│   │   ├── LevelBadge.tsx              # Compact player level badge with animated gold
bar
│   │   ├── SimulationBadge.tsx         # Dev-mode red glowing simulation status badge
│   │   ├── TypewriterText.tsx          # Word-by-word / character-by-character
typewriter
│   │   ├── WaypointPhoto.tsx           # Memory photo container with memory badge
│   │   └── XPFlash.tsx                 # Floating upward-drifting +XP ember notification
│   ├── types/
│   │   └── quest.ts                    # Core domain TypeScript models & schemas
│   └── utils/
│       ├── geo.ts                      # Haversine distance & linear coordinate
interpolation
│       ├── heroCard.ts                 # 1080x1080 Canvas graphic exporter for shareable
cards
│       ├── manuscript.ts               # Segmenter for narrative unrolling
│       ├── svgToPng.ts                 # High-res SVG to raster canvas PNG downloader
│       └── xp.ts                       # Level boundaries, titles, and multiplier logic
└── world-dungeon-backend/
    ├── .env                            # Fastify backend environment config
    ├── .env.example                    # Fastify backend environment template
    ├── .gitignore                      # Node modules & build artifacts exclusions
    ├── package.json                    # Fastify, Zod, dotenv, pino-pretty, tsx
    ├── README.md                       # Backend setup and API manual
    ├── tsconfig.json                   # Strict ES2022/NodeNext TypeScript configuration
    └── src/
        ├── index.ts                    # Fastify server entry, CORS, graceful shutdown
        ├── config.ts                   # Zod-parsed environment configuration (fail-
fast)
        ├── routes/
        │   ├── generateQuest.ts        # POST /api/generate-quest (complete quest
pipeline)
        │   ├── fetchImage.ts           # GET /api/fetch-image (Wikimedia Commons
gateway)
        │   ├── enrichLandmark.ts       # GET /api/enrich (SerpApi landmark enrichment)
        │   └── health.ts               # GET /api/health (Ollama & Overpass ping probe)
        ├── services/
        │   ├── overpass.ts             # Overpass QL OSM query, snapping & scoring
        │   ├── ollama.ts               # Ollama REST interface, gemma2:2b caller &
retrier
        │   ├── wikimedia.ts            # 3-tier Wikipedia/Wikimedia image fetcher +
cache
        │   └── serpapi.ts              # SerpApi web search client with 3s timeout
        ├── prompts/
        │   └── questPrompt.ts          # Regional lore prompt builder with tone auto-
detect
        ├── validators/
```

```
│   └── questSchema.ts            # Zod schemas for input/output verification
└── utils/
    ├── retry.ts                  # Exponential backoff retry utility
    ├── roman.ts                  # Decimal to Roman numeral encoder
    └── sanitize.ts               # Resilient JSON cleaner for LLM output anomalies
```

3. COMPLETE FRONTEND SPECIFICATION

3.1 Screen 1: The Living Breathing Home Screen (`src/app/page.tsx`)

Rather than presenting a static splash page, the Home Screen runs a persistent 60fps canvas engine simulating an atmospheric, living world.

Layered Architecture:

- **Layer 0: Fullscreen HTML5 Particle & Fog Canvas (`z-index: 0`)**:
  - **Fog Ellipses**: 6–8 independent radial gradient ellipses (`rgba(139, 90, 43, 0.08)` to transparent) positioned at 20%, 50%, and 75% screen height. They drift horizontally (± 30px ±30px) and vertically (± 15px±15px) using trigonometric sine functions with individual phase offsets.
  - **Embers & Ash**: 60 floating particles on desktop (automatically throttled to 35 on mobile via touch detection). Colors: `#c9a84c` (gold), `#ff6b35` (orange), `#ff4500` (bright ember, 20% frequency), `#e8e0d0` (white ash, 15% frequency). Particles drift upward, wobble sinusoidally, and feature **interactive cursor repulsion physics**: moving the mouse or dragging a finger within 80px80px pushes particles outward with decaying velocity.
  - **Subliminal Geometric Runes**: 6 geometric, angular SVG vector watermarks floating across the background at 0.03 to 0.06 opacity, executing slow rotations (30–60 seconds per cycle).
- **Layer 1: The Map Tear Geometry (`z-index: 1`)**:
  - A 14-point irregular polygon vector styled like a torn, ancient parchment fragment ($280\text{px} \times 380\text{px}$280px × 380px).
  - Filtered via an SVG `feTurbulence` and `feDisplacementMap` pipeline (`baseFrequency: 0.04` , `numOctaves: 3`) to yield aged, ragged, gold-leaf borders.
  - Layered with an internal dotted grid pattern (30px30px intervals, 0.12 opacity).
  - Oscillates with an organic 4-second pulse animation (`map-tear-pulse`), expanding by 15% 15% when geolocation permission is granted.

- **Layer 2: Atmospheric Dark Vignette (`z-index: 2`)**:
  - Fullscreen pointer-events-disabled gradient: transparent at center (30%30%), darkening to `rgba(10, 10, 15, 0.4)` at 60%60%, and terminating at `rgba(10, 10, 15, 0.85)` at the viewport perimeter.
- **Layer 10: HUD Top Status Bar**:
  - Left: Game engine version `[ REALM MAPPER V1.0 ]`.
  - Integration Badge: Live indicator reading either `● LLM LIVE` (green-gold `#7a9a4c`, animated pulse) when the Fastify backend and Ollama are reachable, or `○ ARCHIVE` when operating in standalone offline mode.
  - Center: Displays `◆ LOCATION LOCKED` when geolocation is approved.
  - Right: Web Audio API ambient drone toggle (3 oscillating visual equalizer bars that activate a 60Hz60Hz deep sine drone when clicked) alongside a real-time `HH:MM` clock.
- **Layer 10: Central Title & RPG Elements**:
  - **Title**: "WORLD DUNGEON" in Cinzel 700 with a 4-stop vertical gradient (`#f0d060` → → `#c9a84c` → → `#8b6914` → → `#c9a84c`) backed by an identical blurred clone acting as an optical gold halo.
  - **Ornamental Divider**: An SVG horizontal rule with flanking lines and a center diamond glyph featuring a `stroke-dashoffset` line-drawing animation.
  - **Stats Pills**: Horizontal indicators showing completed quests (`⚔ X QUESTS`), distance walked (`◆ Xkm WALKED`), and current level (`✦ LVL X`), or `✦ THE FIRST QUEST AWAITS` for new walkers.
  - **Level Badge**: RPG status card displaying level number, title, and an animated gold shimmer bar showing XP progress toward the next threshold.
  - **Hexagonal Gate Action Button**:
    - **State A ("REVEAL THE MAP")**: 6-sided shield geometry with decorative corner brackets, border pulsing, and hover ember bursts. Requests browser GPS.
    - **State B ("CONSULTING...")**: Active during quest generation. Features counter-rotating rune locks (`animate-spin-cw` and `animate-spin-ccw`) and cycles through 7 loading phrases every 2.5 seconds.
    - **State C ("GENERATE QUEST")**: Enabled once GPS coordinates are locked, showing a green beacon dot.
- **Layer 10: Bottom HUD**:
  - Real-time GPS coordinate readout (`XX.XXXX° N YY.YYYY° E`).
  - Cascading double down chevrons (`v v`) indicating depth.

    - "TOUCH GRASS" meta label.

3.2 Screen 2: High-Contrast 1-Second Glancable Walk HUD (`src/app/walk/page.tsx`)

Designed specifically for walking outdoors under sunlight. Readable within 1 second without visual distraction.

- **Top Bar**: Fixed display of Quest Title, active waypoint ratio (`II / IV`), and simulation indicator.
- **Primary Distance Display**:
    - **Sequence Number**: 80px Cinzel gold Roman numeral (e.g., `II / IV`).
    - **Real Place Name**: 20px IM Fell English bold white text.
    - **Remaining Distance**: 48px parchment number formatted strictly as meters (e.g., `47m`).
- **Field Photography Module (`WaypointPhoto.tsx`)**:
    - **Locked State**: 4:3 aspect ratio dark container with `#3d2b1f` background, central gold diamond, and italicized place name.
    - **Unlocked (No User Photo)**: Displays the Wikimedia Commons image (or a camera prompt) with a circular 48px gold camera button calling the native camera interface.
    - **Unlocked (User Photo Captured)**: Displays user's camera photo with vignette overlay, "YOUR MEMORY" badge, retake button, and awards $+30$ XP+30 XP.
- **Proximity Unlock Sequence (< 20 meters< 20 meters)**:
    1. Entire screen flashes gold once via a 150ms CSS animation (`animate-flash-gold`).
    2. Numbers fade out and transition to the unlock screen.
    3. "YOU HAVE ARRIVED" heading renders in 24px Cinzel.
    4. Typewriter reveal of the waypoint's lore fragment at 40ms per word.
    5. Triggers voice narration using the Web Speech API.
    6. Displays "CONTINUE WALKING" (or "CLAIM THE ARTIFACT" on the final waypoint).
- **Bottom Navigation**: Thin horizontal gold progress bar with pulsing "KEEP WALKING" text.

3.3 Screen 3: Relic Vault & Artifact Screen (`src/app/artifact/page.tsx`)

The reward screen presented upon completing all waypoints.

- **Entrance Animation**: 1-second fade-in transition from black.
- **Top Header**: Quest title in large gold Cinzel with horizontal rules.
- **Dungeon Map Vector (`DungeonMapSVG.tsx`)**:

    - Rendered as an inline $340\text{px} \times 340\text{px}$340px × 340px SVG canvas.
    - Filtered with an inline `feTurbulence` and `feDiffuseLighting` parchment noise texture.
    - Dotted gold trace line connecting GPS coordinates.
    - Gold diamond markers (`◆`) positioned at waypoint coordinates, annotated with Roman numerals (`I` , `II` , `III` , `IV`).
    - Mini compass rose anchored in the lower right corner and distance scale bar in the lower left.
- **Burnt Manuscript Component (`BurntManuscript.tsx`)**:
    - Displays the assembled quest scroll (Premise + All Waypoint Lore Fragments + Resolution).
    - Redacted segments are hidden under a burnt texture (`#1a0a00`) with ragged edges, ember boundary glows (`#ff4500`), and a "STORY UNCOVERED" progress bar.
- **Field Scrapbook**: Chronological journal layout showing all waypoint photographs (user-captured or auto-fetched) paired with lore fragments.
- **Dual Exporters**:
    1. **Download Map**: Renders the SVG map to an HTML5 Canvas and downloads it as `dungeon-map.png` .
    2. **Share Hero Card (`heroCard.ts`)**: Synthesizes a high-resolution $1080\text{px} \times 1080\text{px}$ 1080px × 1080px PNG graphic containing the user's field photo, quest title, premise excerpt, walk statistics, walker level, and compass insignia.
- **Footer & Progression**: Displays quest stats, date, forge provenance badge (`LIVE REALM FORGE · gemma2:2b` or `ANCIENT ARCHIVE`), and a "NEW QUEST" reset button.

4. CORE UTILITIES, HOOKS & STATE MACHINE

4.1 State Management (`QuestContext.tsx`)

Operates via React Context and provides state synchronization with `localStorage` :

- Key `wd_session` : Serializes `currentQuest` , `currentWaypointIndex` , `unlockedWaypoints` , `walkStats` , `gpxTrace` , `isComplete` , and `questMeta` .
- Key `wd_player` : Serializes `PlayerLevel` data (`level` , `xp` , `xpToNext` , `title` , `totalQuestsCompleted` , `totalDistanceMeters`).
- **Automatic Rehydration**: Resuming the app after a reload or closing the browser restores the exact quest step, walked distance, and photos taken.

4.2 Geolocation & Simulation Engine (`useGeolocation.ts`)

- **Production Mode**: Calls `navigator.geolocation.watchPosition` with `enableHighAccuracy: true`, `maximumAge: 5000`, and `timeout: 10000`.
- **Desktop / Development Simulation Mode**:
  - Detects when running in development or when GPS permissions are unavailable.
  - Calculates an offset coordinate 200 meters200 meters away from the current target waypoint.
  - Smoothly interpolates coordinates over 30 seconds toward the destination using Haversine calculation.
  - Automatically trips the proximity unlock ($< 20\text{m}$< 20m) at the 30-second mark to allow full testing of animations, sound, and transitions on desktop.
  - Surfaces a glowing red `SIMULATION MODE` badge on the Walk HUD.

4.3 Voice Narration (`useVoiceNarration.ts`)

- Interacts with `window.speechSynthesis`.
- Voice Selection Priority:
  1. Any voice containing "deep" in its name.
  2. Any voice containing "male" in its name.
  3. First available English voice (`en-*`).
- Playback Parameters: `rate: 0.82`, `pitch: 0.65`, `volume: 1.0`.
- Staggered Execution: Speaks the `unlockCue` ("You have reached the threshold"), pauses for 800ms800ms, then speaks the full `loreFragment` at `rate: 0.78` and `pitch: 0.62`. Degrades silently if speech synthesis is unavailable.

4.4 Camera Capture (`useCamera.ts`)

- Dynamically generates an `<input type="file" accept="image/*" capture="environment">` element.
- Directly triggers the device's rear camera hardware on mobile browsers.
- Reads the resulting photo as a Base64 Data URL and saves it to the waypoint state and `localStorage`.

4.5 XP & Leveling Engine (`xp.ts`)

- **Level Titles (1–10)**:

1. *The Wanderer* (0–99 XP)
2. *The Curious* (100–249 XP)
3. *The Initiated* (250–499 XP)
4. *The Seeker* (500–899 XP)
5. *The Cartographer* (900–1399 XP)
6. *The Keeper* (1400–1999 XP)
7. *The Unbound* (2000–2699 XP)
8. *The Ancient* (2700–3499 XP)
9. *The Forsaken* (3500–4499 XP)
10. *The Dungeon Master* (4500+ XP)

- **XP Rewards**:
  - Quest Completion: $+50\ \mathrm{XP}$+50 XP
  - Waypoint Unlocked: $+20\ \mathrm{XP}$+20 XP each
  - User Photo Captured: $+30\ \mathrm{XP}$+30 XP each
  - Distance Walked: $+1\ \mathrm{XP}$+1 XP per $100\ \mathrm{meters}$100 meters
  - All Waypoints Cleared Bonus: $+25\ \mathrm{XP}$+25 XP
  - All User Photos Taken Bonus: $+50\ \mathrm{XP}$+50 XP
- **Visual Effects**: Triggers the floating `XPFlash` component (`+30 XP` in `#ff4500`) and pulses the `LevelBadge` with an ember glow for 3 seconds upon level-up.

5. COMPLETE BACKEND SPECIFICATION (`world-dungeon-backend`)

5.1 Architecture & Principles

A standalone Node.js 18+ service running Fastify and TypeScript. It serves as the procedural generation engine, converting raw GPS coordinates into structured quest JSON.

- **Zero Cloud Dependence**: The default stack (Overpass API + local Ollama + Wikimedia Commons) costs $\$0.00$$0.00 per quest and sends no user coordinates to cloud providers.
- **Fail-Fast Validation**: Configuration is parsed through Zod on boot; invalid or missing settings immediately halt the process.

5.2 OpenStreetMap Overpass Service (`overpass.ts`)

Executes an Overpass QL query searching for landmarks within a radius (default $1000\mathrm{m}$1000m):

```
overpass
[out:json][timeout:25];
(
  node["name"]
["amenity"~"place_of_worship|park|fountain|monument|library|theatre|cinema|hospital|police|
fire_station"](around:${radius},${lat},${lng});
  node["name"]["historic"](around:${radius},${lat},${lng});
  node["name"]["tourism"~"attraction|artwork|museum|viewpoint|ruins"]
(around:${radius},${lat},${lng});
  node["name"]["leisure"~"park|garden|nature_reserve|pitch"]
(around:${radius},${lat},${lng});
  way["name"]["highway"~"footway|path|steps|pedestrian"](around:${radius},${lat},${lng});
);
out body;
>;
out skel qt;
```

- **Post-Processing Pipeline**:
  1. **Deduplication**: Calculates Levenshtein edit distance between place names; entries within a distance ≤ 3≤ 3 are filtered out (e.g., prevents "Gate 1", "Gate 2").
  2. **Footway Snapping**: Scans nearby pedestrian paths (`footway`, `path`, `steps`, `pedestrian`). If no path exists within 50m50m, marks the landmark with `mayBeInaccessible: true`.
  3. **Scoring Engine**:
     - $+3$+3 points for `historic` tag
     - $+3$+3 points for `heritage` tag
     - $+2$+2 points for `amenity=place_of_worship`
     - $+2$+2 points for `tourism=ruins` or `tourism=monument`
     - $+1$+1 point for `description` or `wikipedia` tags
     - -2−2 points if `mayBeInaccessible: true`
  4. **Radius Widening**: Automatically re-queries at 2000m2000m if fewer than 4 valid candidates are found within the initial radius.

5.3 Local LLM Service (`ollama.ts`)

- Interacts with local Ollama (`http://localhost:11434`) running `gemma2:2b`.
- Generation parameters: `temperature: 0.85`, `top_p: 0.92`, `top_k: 40`, `repeat_penalty: 1.1`, `num_predict: 1200`.
- **Sanitization (`sanitize.ts`)**: Cleans Markdown code fences (```` ```json ````), strips leading/trailing preambles by extracting the outermost `{` and `}`, removes zero-width/BOM artifacts, and

corrects trailing commas before `}` or `]`.

- **Validation & Retries**: Validates output against `QuestSchema`. If validation fails, retries once with a strict formatting reminder.

5.4 Procedural Prompt Builder (`questPrompt.ts`)

- **Tone Auto-Detection**:
  - Landmark contains Islamic history/mosque → `folkloric` (Mughal folklore).
  - Landmark contains fort/ruins/castle → `mythic`.
  - Local time between 18:00 and 23:59 → `dark`.
  - Default → `eerie`.
- **Prompt Directives**:
  - Mandates exactly 4 waypoints chosen strictly from the OSM landmark list.
  - Requires each lore fragment (80–150 words80–150 words) to mention the real place name at least twice.
  - Rejects generic European fantasy tropes in favor of regional history and urban legends.

5.5 Wikimedia Commons Service (`wikimedia.ts`)

Fetches images concurrently for all waypoints using a 3-tier fallback strategy:

1. **Tier 1 (Wikipedia Summary API)**: Queries `https://en.wikipedia.org/api/rest_v1/page/summary/{placeName}` to extract `thumbnail.source`.
2. **Tier 2 (Commons Direct File)**: Queries `https://commons.wikimedia.org/w/api.php` for exact file title matches.
3. **Tier 3 (Commons Full-Text Search)**: Searches Wikimedia Commons namespace 6 for place names.

- Features an in-memory `Map<string, string | null>` cache to prevent duplicate lookups.

5.6 SerpApi Landmark Enrichment (`serpapi.ts`)

- Optional service activated when `SERPAPI_KEY` is present in `.env`.
- Executes Google historical searches in parallel with a strict 3-second timeout.
- Injects contextual search snippets directly into the prompt (`[REAL CONTEXT: "..."]`) to ground LLM-generated lore in historical facts. Degrades silently if the key is absent or a call times out.

6. ENDPOINT INTEGRATION & GATEWAY ARCHITECTURE

To support development and demo environments, the Next.js API routes act as an intelligent gateway to the Fastify backend:

```
[Client PWA]
     │
     ▼
[Next.js API Gateway (Port 3000)]
     │
     ├──▶ POST /api/generate-quest ──▶ Proxies to Fastify (3001) ──▶ Overpass + Ollama + Wikimedia
     │                                 └─ (Fallback: Returns Lodi Garden archive if backend is offline)
     │
     ├──▶ GET  /api/health         ──▶ Proxies to Fastify (3001) ──▶ Pings Ollama, Overpass, SerpApi
     │                                 └─ (Fallback: Returns degraded status with setup instructions)
     │
     ├──▶ GET  /api/fetch-place-img──▶ Proxies to Fastify (3001) ──▶ Wikimedia cache
     │                                 └─ (Fallback: Direct Wikipedia API lookup)
     │
     └──▶ GET  /api/enrich         ──▶ Proxies to Fastify (3001) ──▶ SerpApi client
```

Gateway Endpoints:

1. **`POST /api/generate-quest`**:
   - Accepts `{ lat, lng, radius?, tone? }`.
   - Forwards requests to Fastify with a 90-second timeout.
   - On success, returns `{ success: true, quest: Quest, meta: { ...meta, source: "live-backend" } }`.
   - If the backend is unreachable, falls back to the curated 4-waypoint *Forgotten Pact of Lodi Garden* quest.
2. **`GET /api/health`**:
   - Probes Fastify at `http://localhost:3001/api/health` with a 2.5-second timeout.
   - Returns connection status for Ollama (`gemma2:2b`), Overpass API, and SerpApi.
   - If unreachable, returns `{ status: "degraded", backendConnected: false, mode: "archive-fallback" }`.

3. `GET /api/fetch-place-image?place=...`:
    - Proxies to Fastify's `/api/fetch-image`, with fallback to direct client-side Wikipedia resolution.
4. `GET /api/enrich?place=...&city=...`:
    - Proxies to Fastify's `/api/enrich` to retrieve real-world historical snippets.

7. VERIFICATION MATRIX & BUILD AUDIT

Component	Target System	Tool / Command	Result	Notes
Frontend Type Check	Next.js 14 / TypeScript	`npx tsc --noEmit`	**Pass (0 Errors)**	Strict mode enabled. Root tsconfig isolates backend.
Backend Type Check	Fastify / TypeScript	`npx tsc --noEmit`	**Pass (0 Errors)**	NodeNext/ES2022 module resolution verified.
Production Build	Next.js App Router	`npm run build`	**Pass (10/10 Routes)**	Generated all static pages and dynamic API routes cleanly.
Backend Boot Probe	Fastify Engine	`npm run dev` (`tsx`)	**Pass**	Bound to `http://0.0.0.0:3001` with CORS enabled for port 3000.
Health API Route	Next.js Gateway	`GET /api/health`	**Pass (200 OK)**	Correctly aggregates and reports service availability.

8. ENVIRONMENT SETUP & EXECUTION GUIDE

8.1 Prerequisites

- **Node.js**: 18.17+ or 20+
- **Ollama**: Download and install from ollama.com.
- **Model**: Pull the Gemma model:

```
bash
ollama pull gemma2:2b
```

8.2 Starting the Backend Engine

```
bash
cd d:\World-dungeon\world-dungeon-backend
npm install
npm run dev
```

Backend runs on `http://localhost:3001`.

8.3 Starting the Frontend PWA

In a separate terminal:

```
bash
cd d:\World-dungeon
npm install
npm run dev
```

Frontend runs on `http://localhost:3000`.

8.4 Verification

1. Open `http://localhost:3000` in your browser.
2. Confirm the top status bar displays the green-gold `• LLM LIVE` badge.
3. Click **REVEAL THE MAP** to enable geolocation.
4. Click **GENERATE QUEST** to run procedural generation via Overpass and Ollama (or test in desktop simulation mode).
5. Walk through the waypoints to test proximity unlocks, voice narration, camera capture, and final artifact generation.